

Image alignment

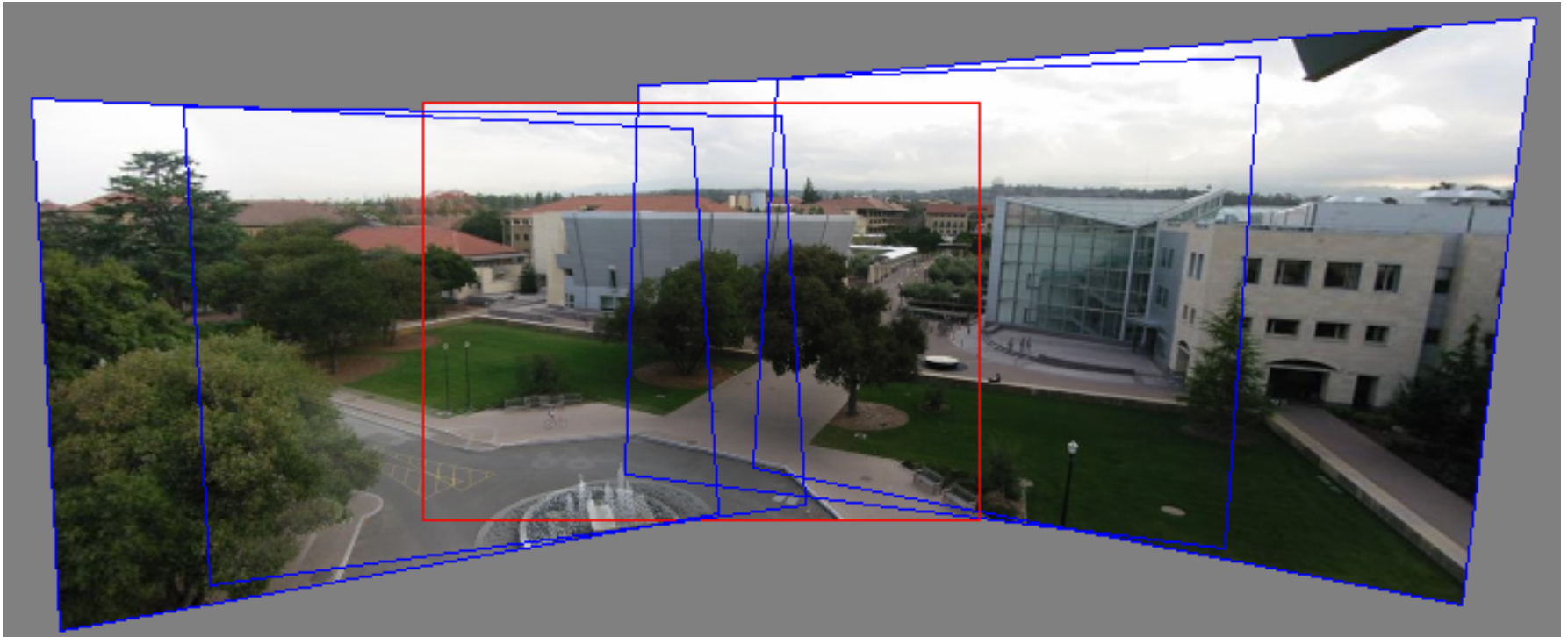
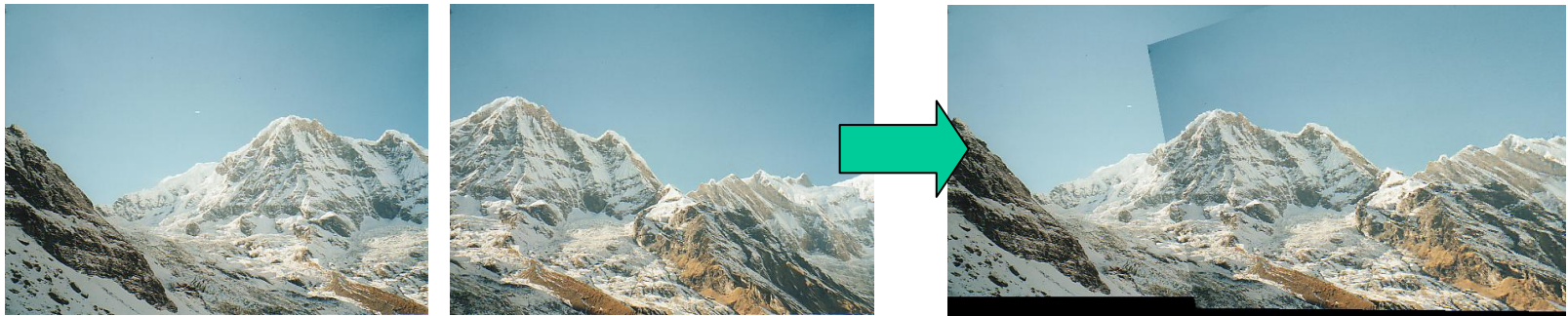


Image alignment: Motivation

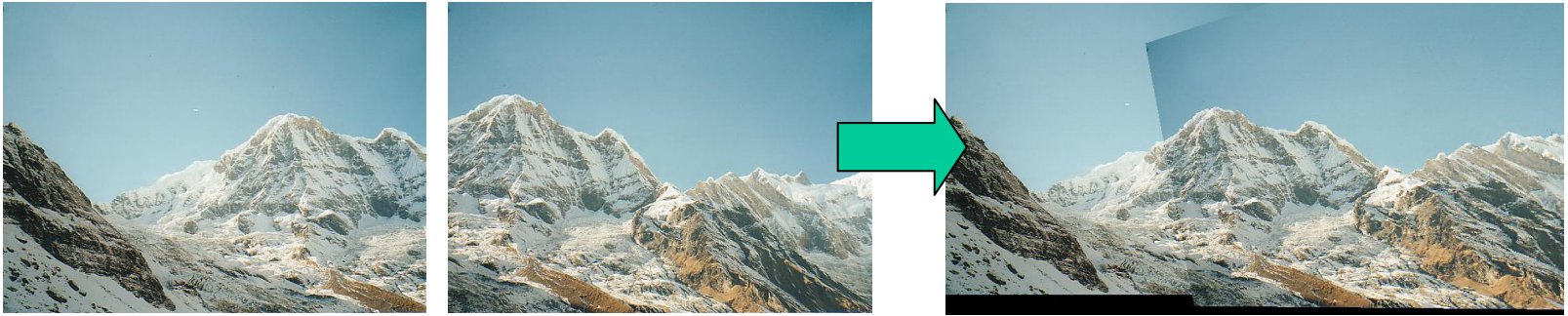


Panorama stitching



Recognition
of object
instances

Image alignment: Challenges

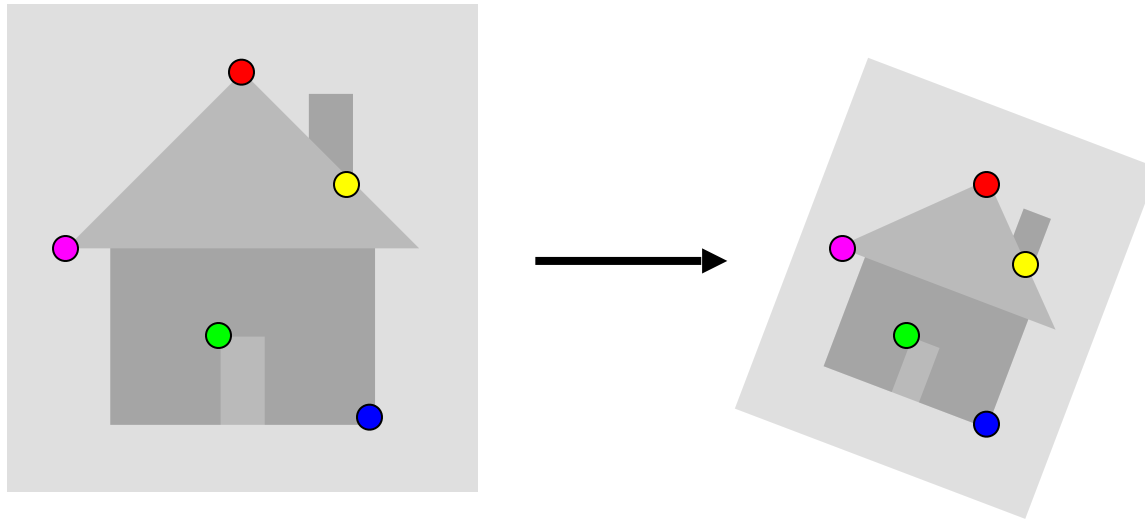


Small degree of overlap



Occlusion,
clutter

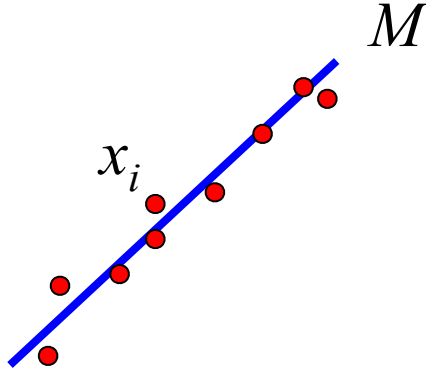
Image alignment



- Two broad approaches:
 - Direct (pixel-based) alignment
 - Search for alignment where most pixels agree
 - Feature-based alignment
 - Search for alignment where *extracted features* agree
 - Can be verified using pixel-based alignment

Alignment as fitting

- Last lecture: fitting a model to features in one image

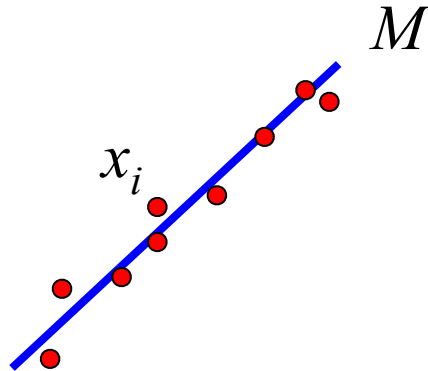


Find model M that minimizes

$$\sum_i \text{residual}(x_i, M)$$

Alignment as fitting

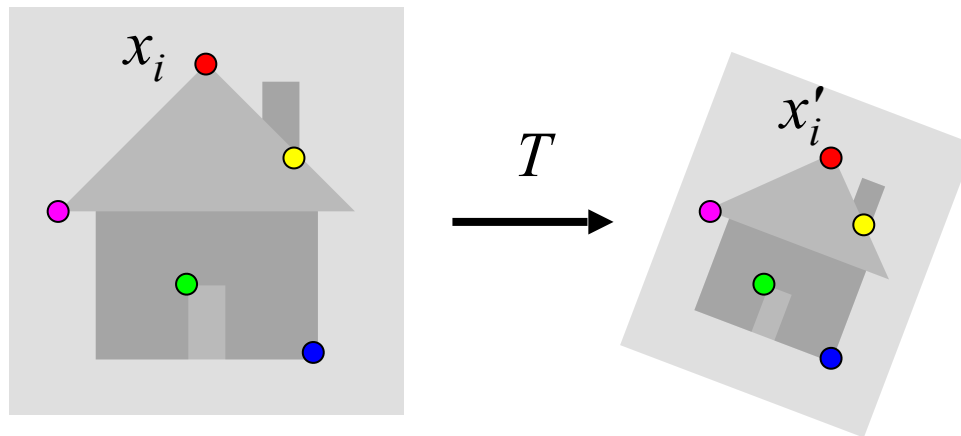
- Last lecture: fitting a model to features in one image



Find model M that minimizes

$$\sum_i \text{residual}(x_i, M)$$

- Alignment: fitting a model to a transformation between pairs of features (*matches*) in two images



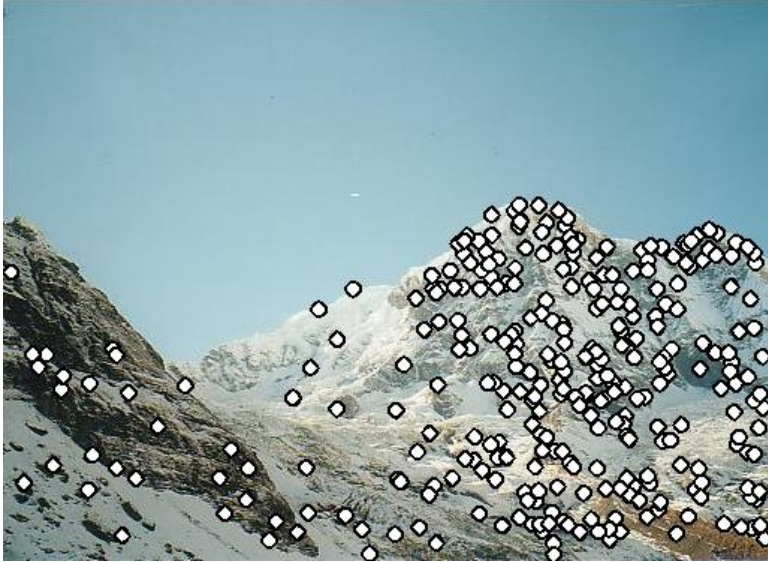
Find transformation T that minimizes

$$\sum_i \text{residual}(T(x_i), x'_i)$$

Feature-based alignment outline

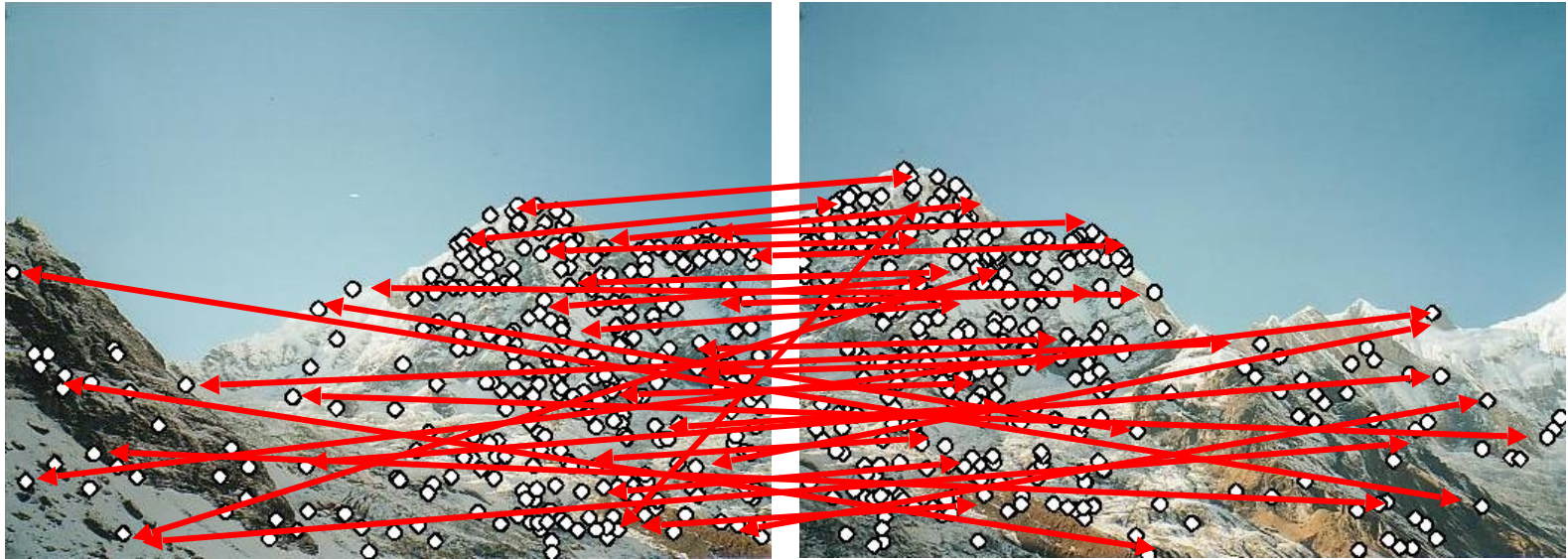


Feature-based alignment outline



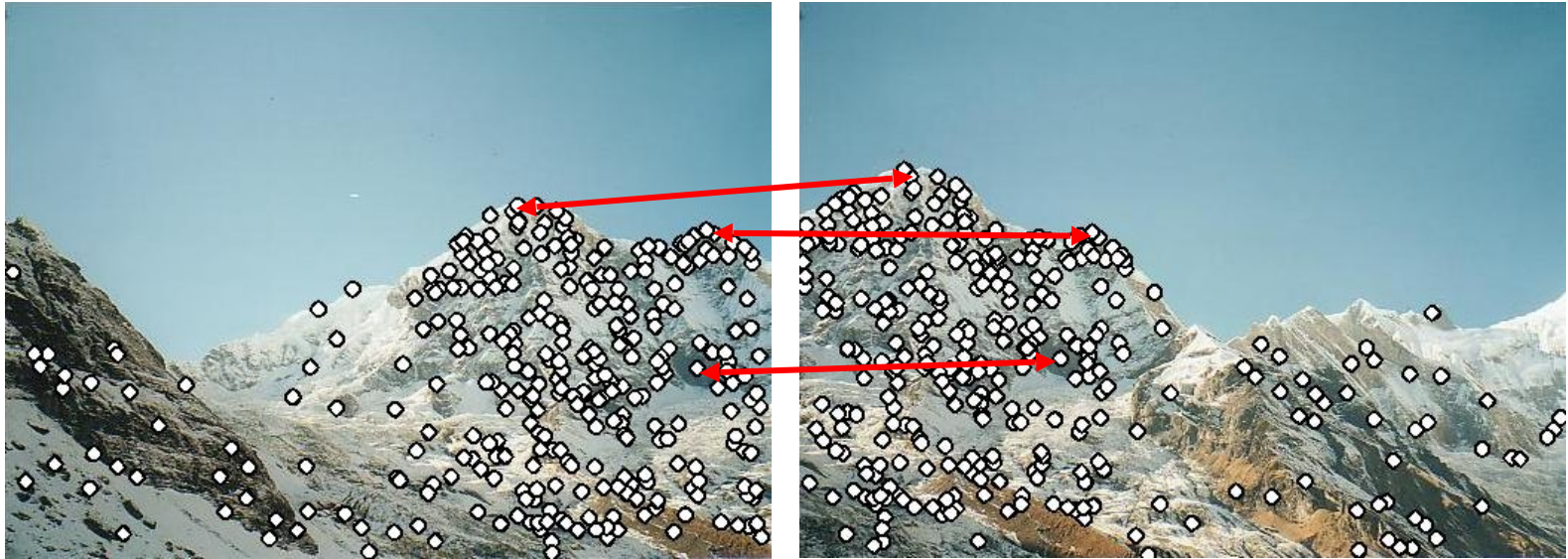
- Extract features

Feature-based alignment outline



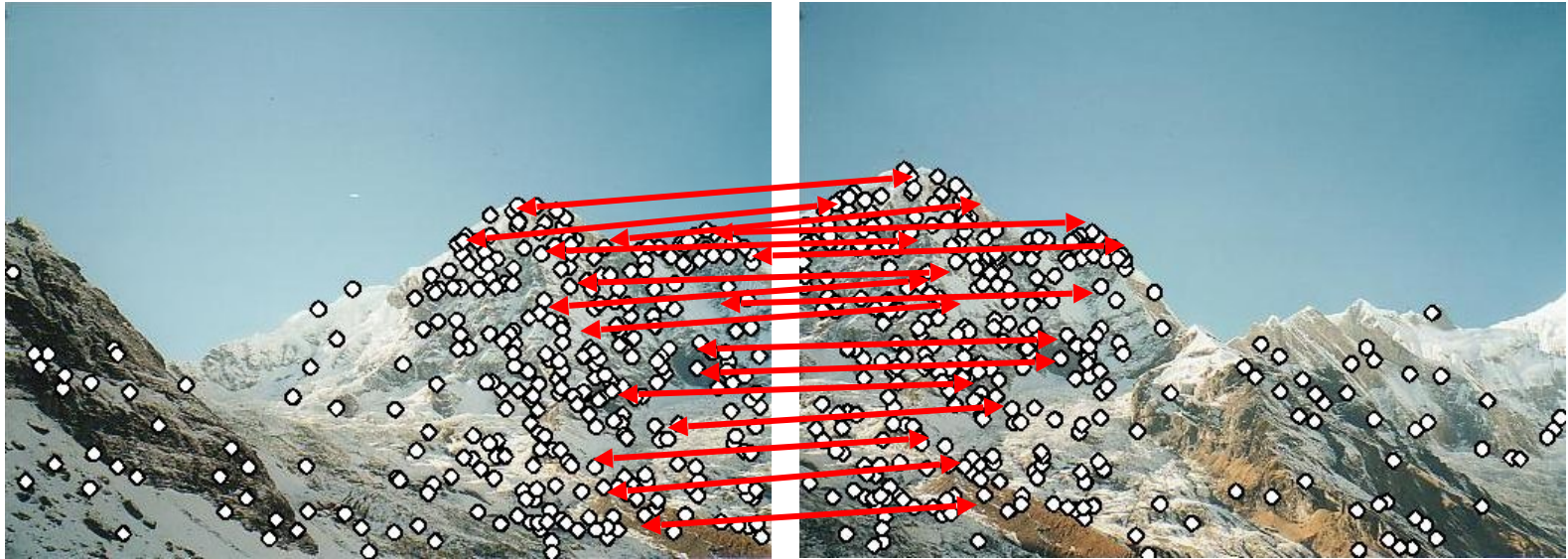
- Extract features
- Compute *putative matches*

Feature-based alignment outline



- Extract features
- Compute *putative matches*
- Loop:
 - *Hypothesize* transformation T (small group of putative matches that are related by T)

Feature-based alignment outline



- Extract features
- Compute *putative matches*
- Loop:
 - *Hypothesize* transformation T (small group of putative matches that are related by T)
 - *Verify* transformation (search for other matches consistent with T)

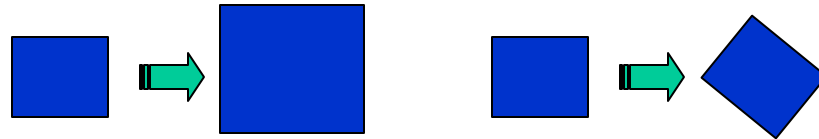
Feature-based alignment outline



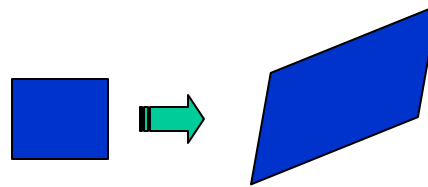
- Extract features
- Compute *putative matches*
- Loop:
 - *Hypothesize* transformation T (small group of putative matches that are related by T)
 - *Verify* transformation (search for other matches consistent with T)

2D transformation models

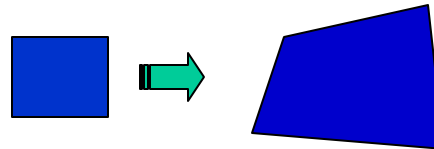
- Similarity
(translation, scale, rotation)



- Affine



- Projective
(homography)



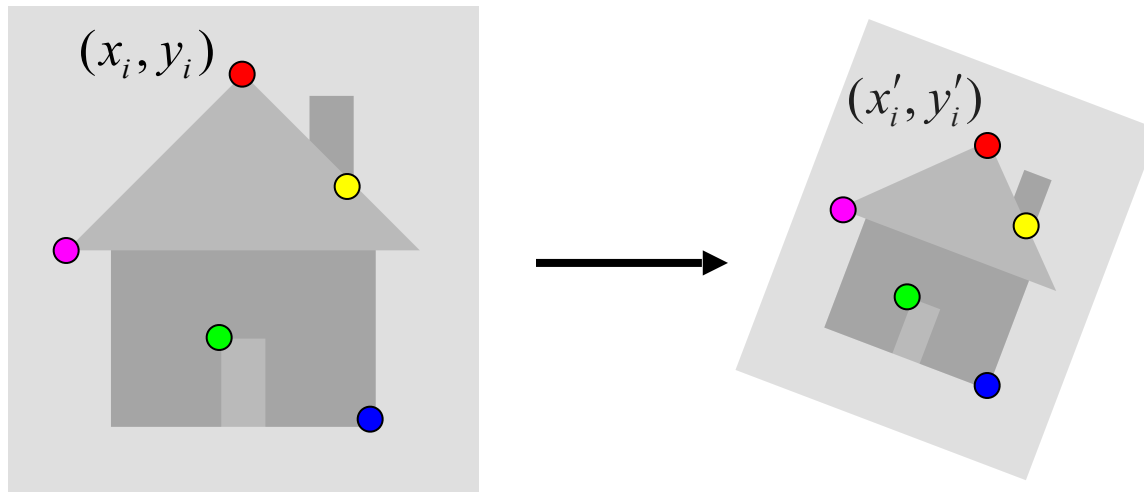
Let's start with affine transformations

- Simple fitting procedure (linear least squares)
- Approximates viewpoint changes for roughly planar objects and roughly orthographic cameras
- Can be used to initialize fitting for more complex models



Fitting an affine transformation

- Assume we know the correspondences, how do we get the transformation?



$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

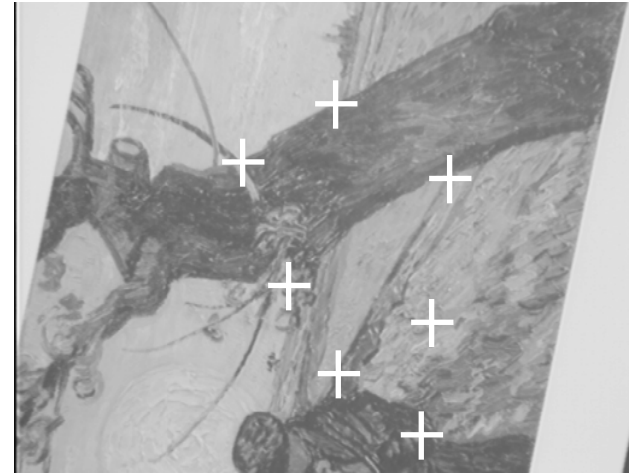
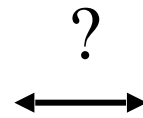
$$\begin{bmatrix} x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \\ \boxed{?} & & & & & \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \boxed{?} \\ x'_i \\ y'_i \\ \boxed{?} \end{bmatrix}$$

Fitting an affine transformation

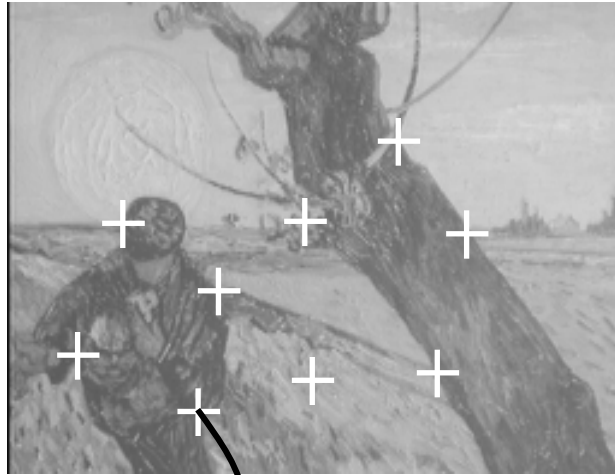
$$\begin{bmatrix} & & \boxed{?} & & & \\ x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \\ & & \boxed{?} & & & \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \boxed{?} \\ x'_i \\ y'_i \\ \boxed{?} \end{bmatrix}$$

- Linear system with six unknowns
- Each match gives us two linearly independent equations: need at least three to solve for the transformation parameters

What if we don't know the correspondences?

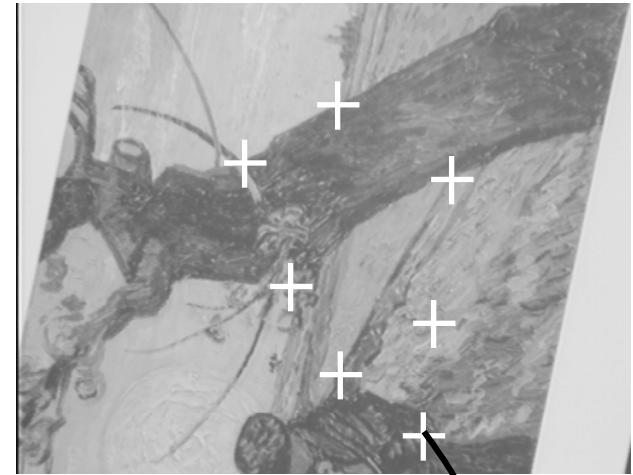


What if we don't know the correspondences?



?

↔



feature
descriptor

?

=

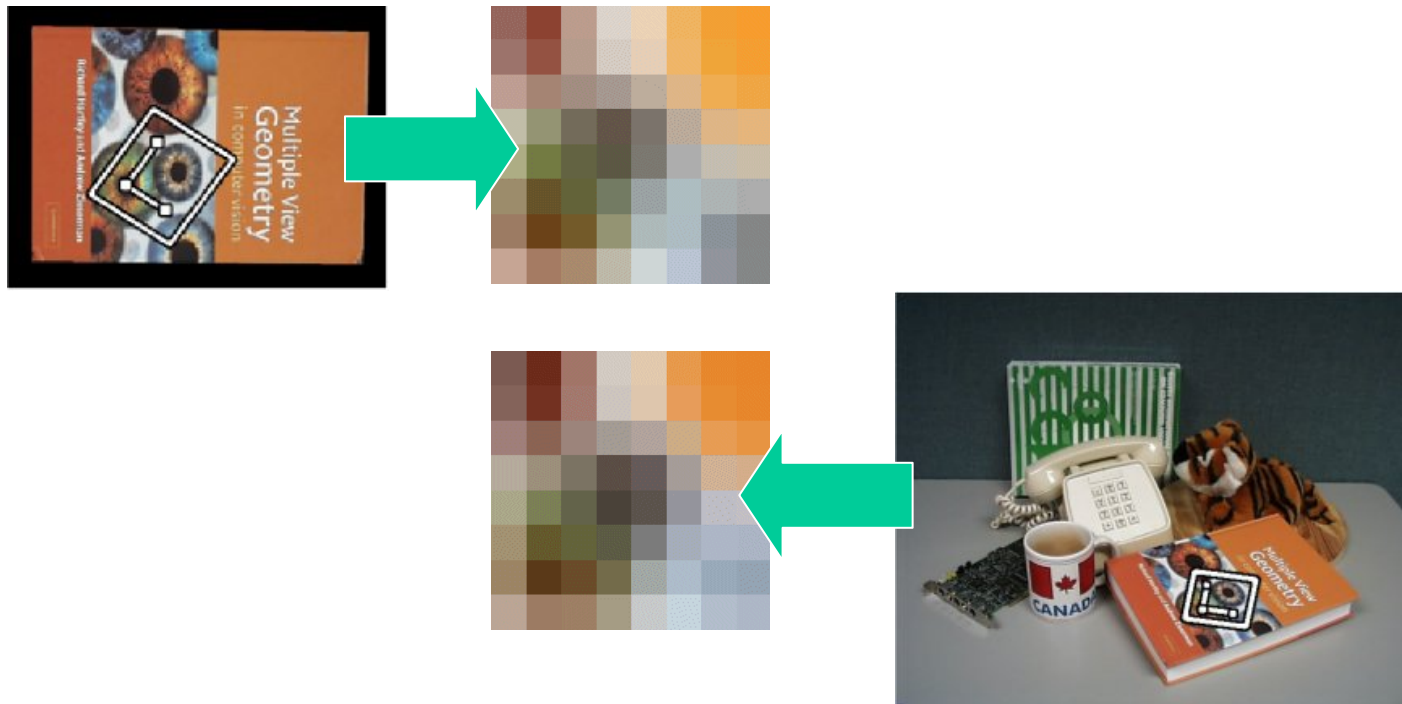


feature
descriptor

- Need to compare *feature descriptors* of local patches surrounding interest points

Feature descriptors

- Assuming the patches are already normalized (i.e., the local effect of the geometric transformation is factored out), how do we compute their similarity?
- Want invariance to intensity changes, noise, perceptually insignificant changes of the pixel pattern



Feature descriptors

- Simplest descriptor: vector of raw intensity values
- How to compare two such vectors?
 - Sum of squared differences (SSD)

$$\text{SSD}(u, v) = \sum_i (u_i - v_i)^2$$

– Not invariant to intensity change

- Normalized correlation

$$\rho(u, v) = \frac{\sum_i (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\left(\sum_j (u_j - \bar{u})^2 \right) \left(\sum_j (v_j - \bar{v})^2 \right)}}$$

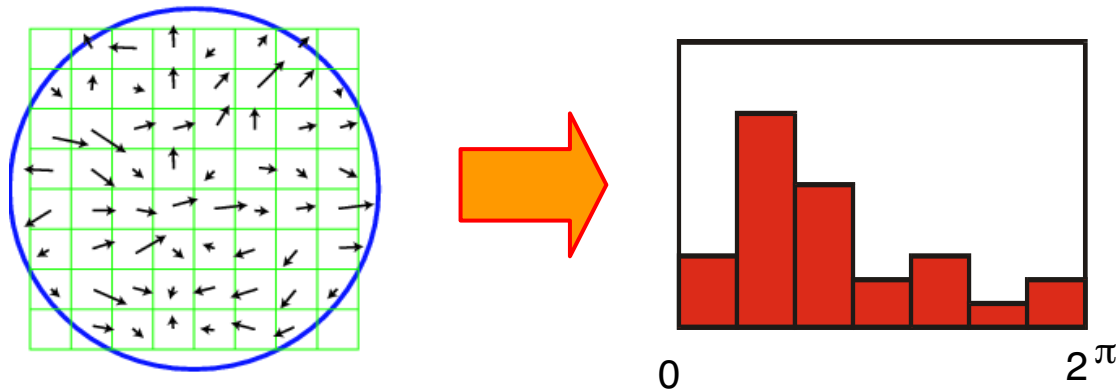
– Invariant to affine intensity change

Feature descriptors

- Disadvantage of patches as descriptors:
 - Small shifts can affect matching score a lot

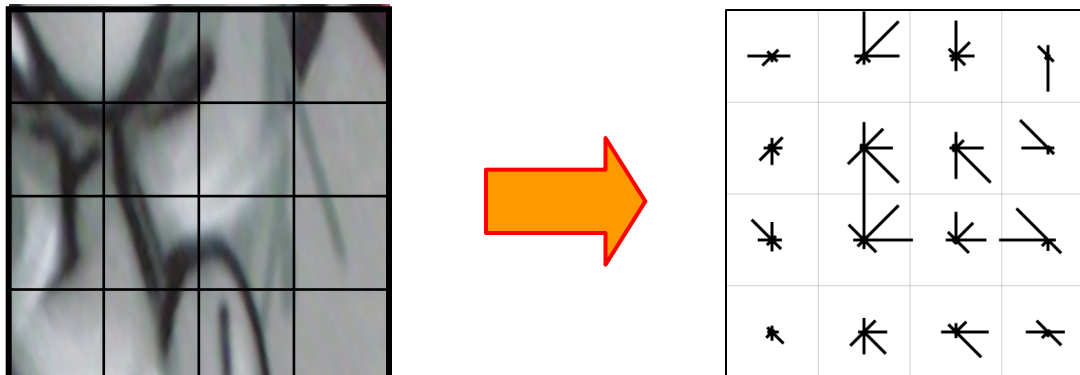


- Solution: histograms



Feature descriptors: SIFT

- Descriptor computation:
 - Divide patch into 4x4 sub-patches
 - Compute histogram of gradient orientations (8 reference angles) inside each sub-patch
 - Resulting descriptor: $4 \times 4 \times 8 = 128$ dimensions



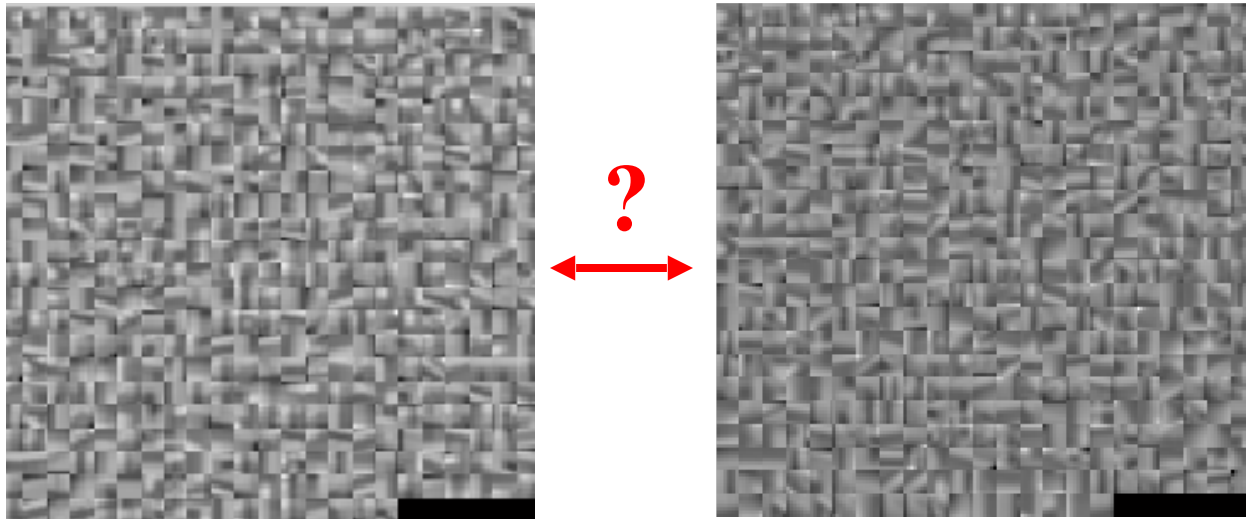
David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#) *IJCV* 60 (2), pp. 91-110, 2004.

Feature descriptors: SIFT

- Descriptor computation:
 - Divide patch into 4x4 sub-patches
 - Compute histogram of gradient orientations (8 reference angles) inside each sub-patch
 - Resulting descriptor: $4 \times 4 \times 8 = 128$ dimensions
- Advantage over raw vectors of pixel values
 - Gradients less sensitive to illumination change
 - “Subdivide and disorder” strategy achieves robustness to small shifts, but still preserves some spatial information

Feature matching

- Generating *putative matches*: for each patch in one image, find a short list of patches in the other image that could match it based solely on appearance

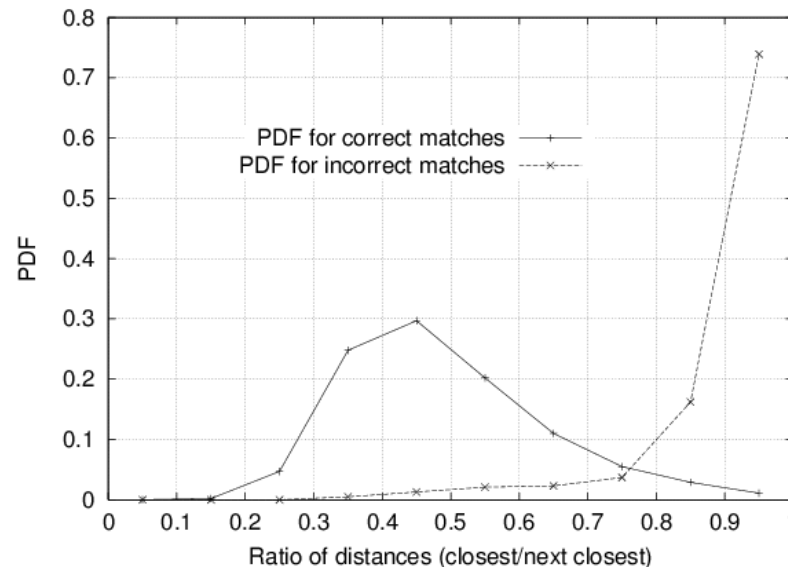


Feature matching

- Generating *putative matches*: for each patch in one image, find a short list of patches in the other image that could match it based solely on appearance
 - Exhaustive search
 - For each feature in one image, compute the distance to *all* features in the other image and find the “closest” ones (threshold or fixed number of top matches)
 - Fast approximate nearest neighbor search
 - Hierarchical spatial data structures (kd-trees, vocabulary trees)
 - Hashing

Feature space outlier rejection

- How can we tell which putative matches are more reliable?
- Heuristic: compare distance of **nearest** neighbor to that of **second** nearest neighbor
 - Ratio will be high for features that are not distinctive
 - Threshold of 0.8 provides good separation



David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#) *IJCV* 60 (2), pp. 91-110, 2004.

Dealing with outliers

- The set of putative matches still contains a very high percentage of outliers
- How do we fit a geometric transformation to a small subset of all possible matches?
- Possible strategies:
 - RANSAC
 - Incremental alignment
 - Hough transform
 - Hashing

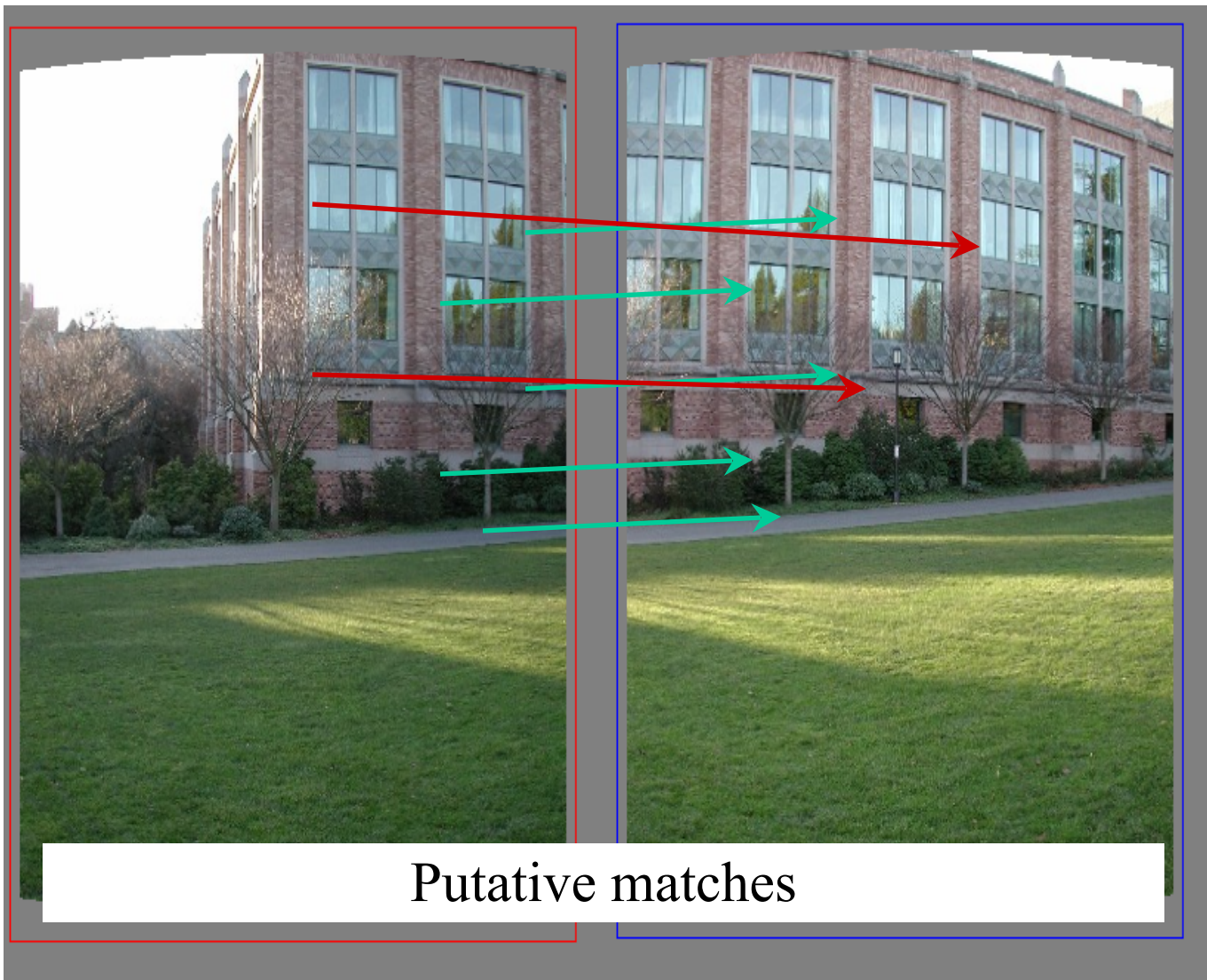
Strategy 1: RANSAC

RANSAC loop:

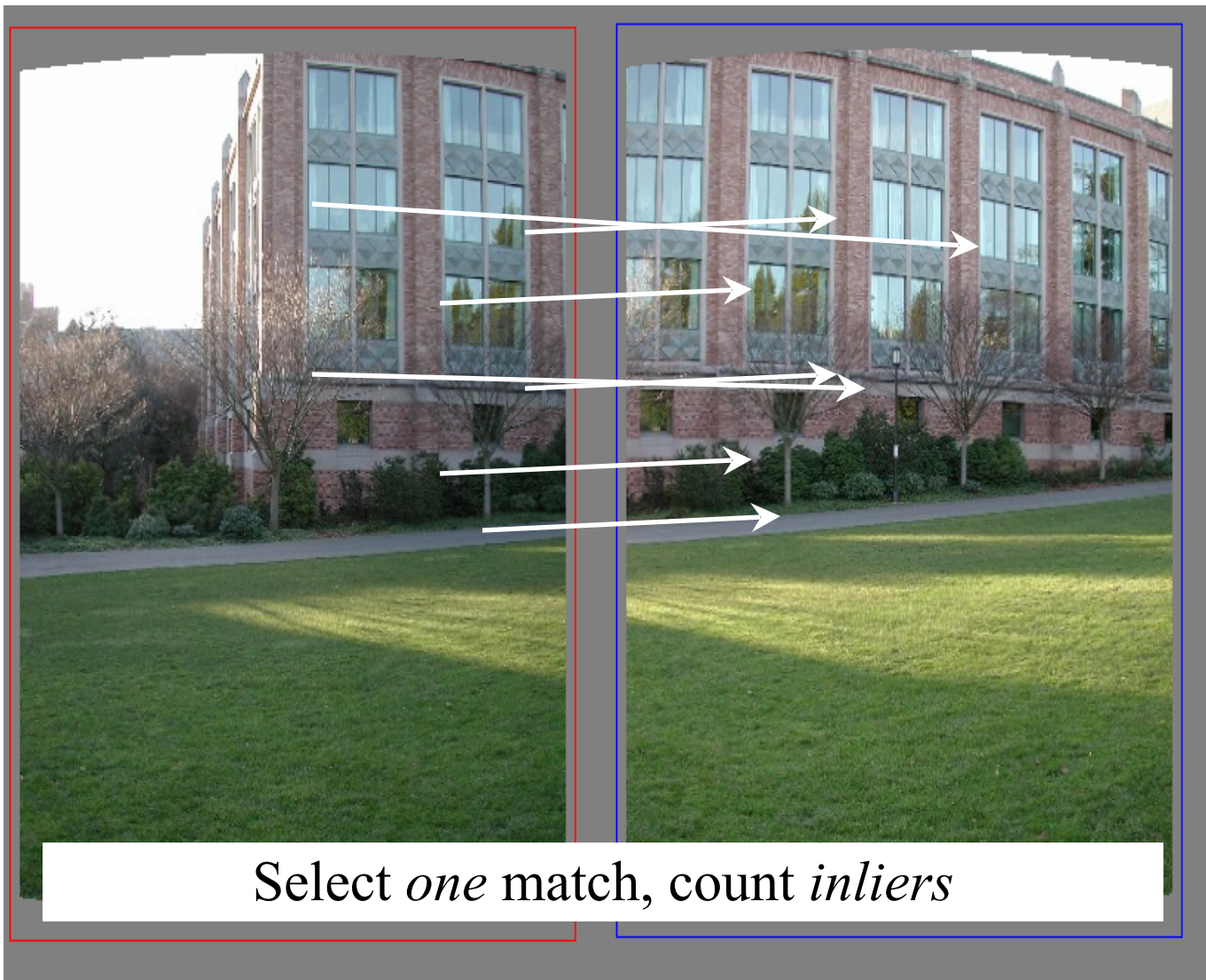
1. Randomly select a *seed group* of matches
2. Compute transformation from seed group
3. Find *inliers* to this transformation
4. If the number of inliers is sufficiently large, re-compute least-squares estimate of transformation on all of the inliers

Keep the transformation with the largest number of inliers

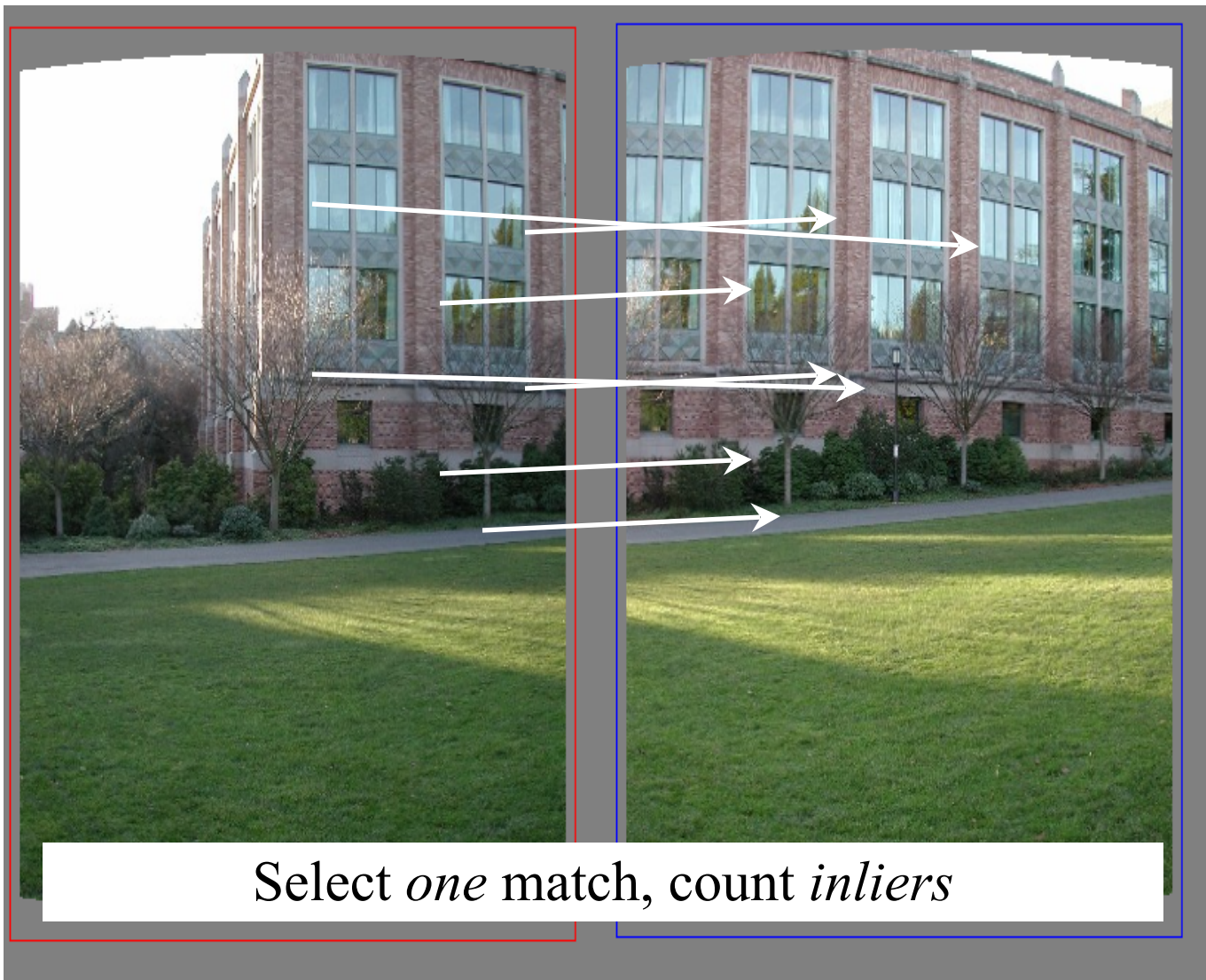
RANSAC example: Translation



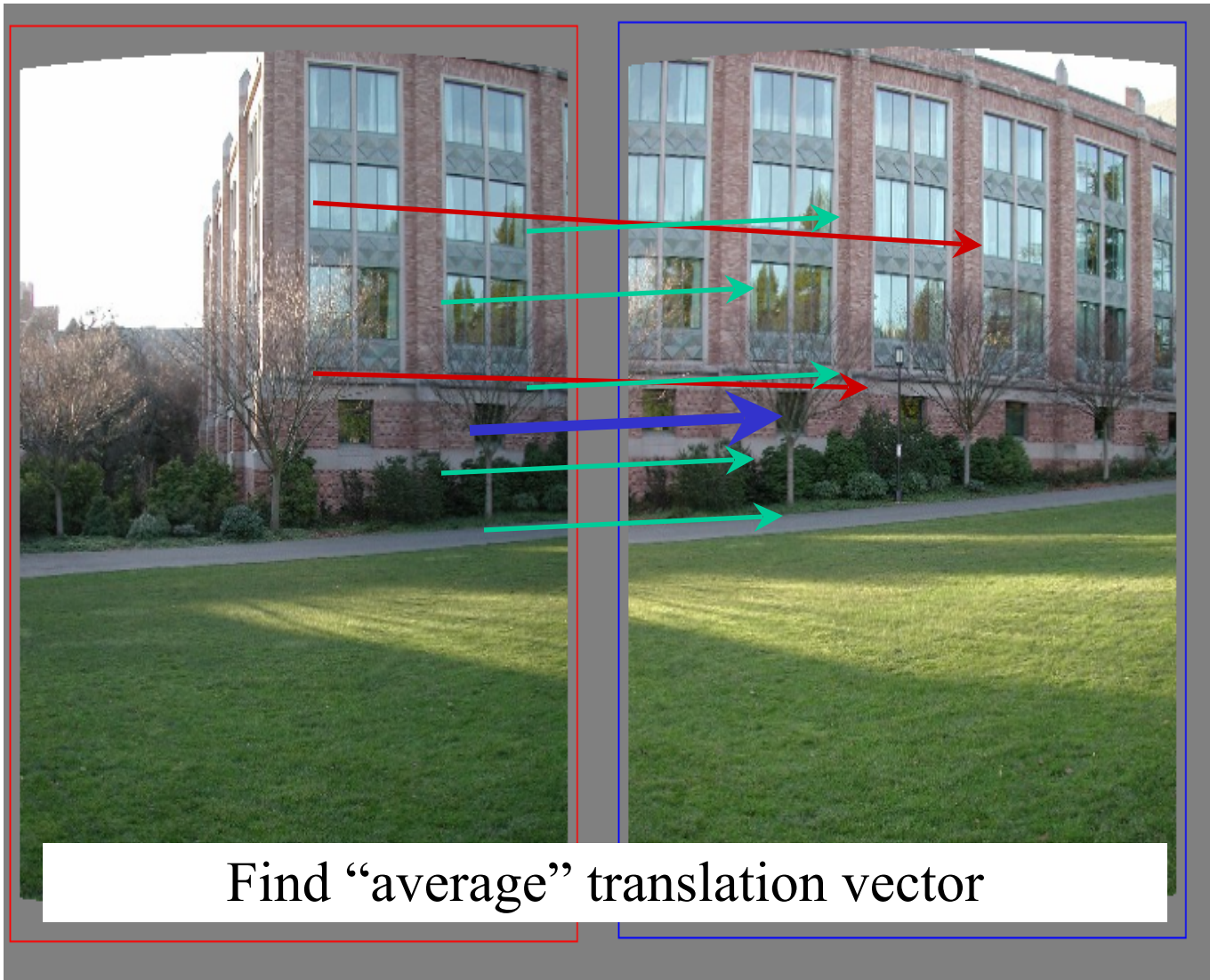
RANSAC example: Translation



RANSAC example: Translation



RANSAC example: Translation



Problem with RANSAC

- In many practical situations, the percentage of outliers (incorrect putative matches) is often very high (90% or above)
- Alternative strategy: restrict search space by using strong locality constraints on seed groups and inliers
 - Incremental alignment

Strategy 2: Incremental alignment

- Take advantage of strong locality constraints: only pick close-by matches to start with, and gradually add more matches in the same neighborhood

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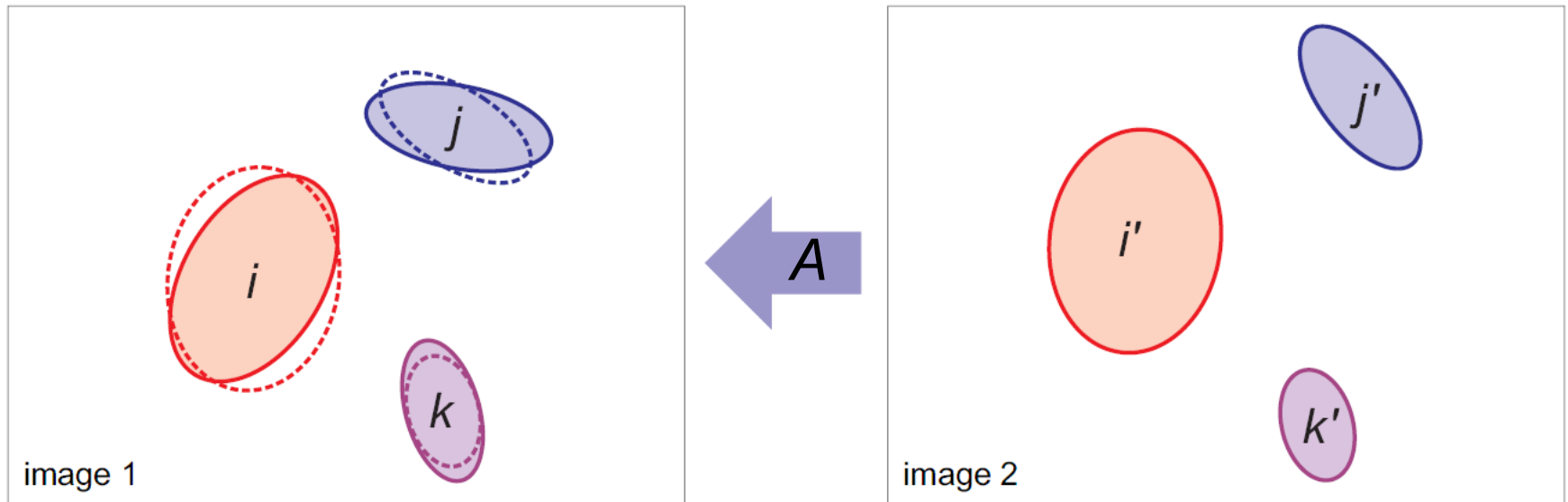
Incremental alignment: Details



Generating seed groups:

- Identify triples of neighboring features (i, j, k) in first image
- Find all triples (i', j', k') in the second image such that i' (resp. j', k') is a putative match of i (resp. j, k), and j', k' are neighbors of i'

Incremental alignment: Details

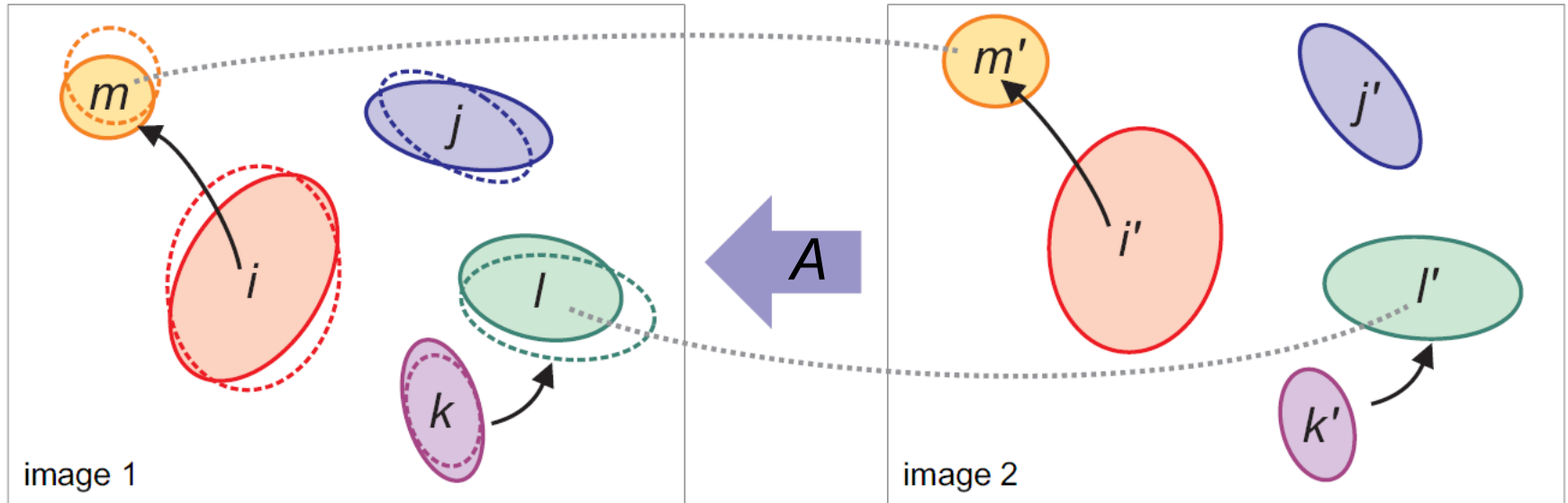


Beginning with each seed triple, repeat:

- Estimate the aligning transformation between corresponding features in current group of matches
- Grow the group by adding other consistent matches in the neighborhood

Until the transformation is no longer consistent
or no more matches can be found

Incremental alignment: Details

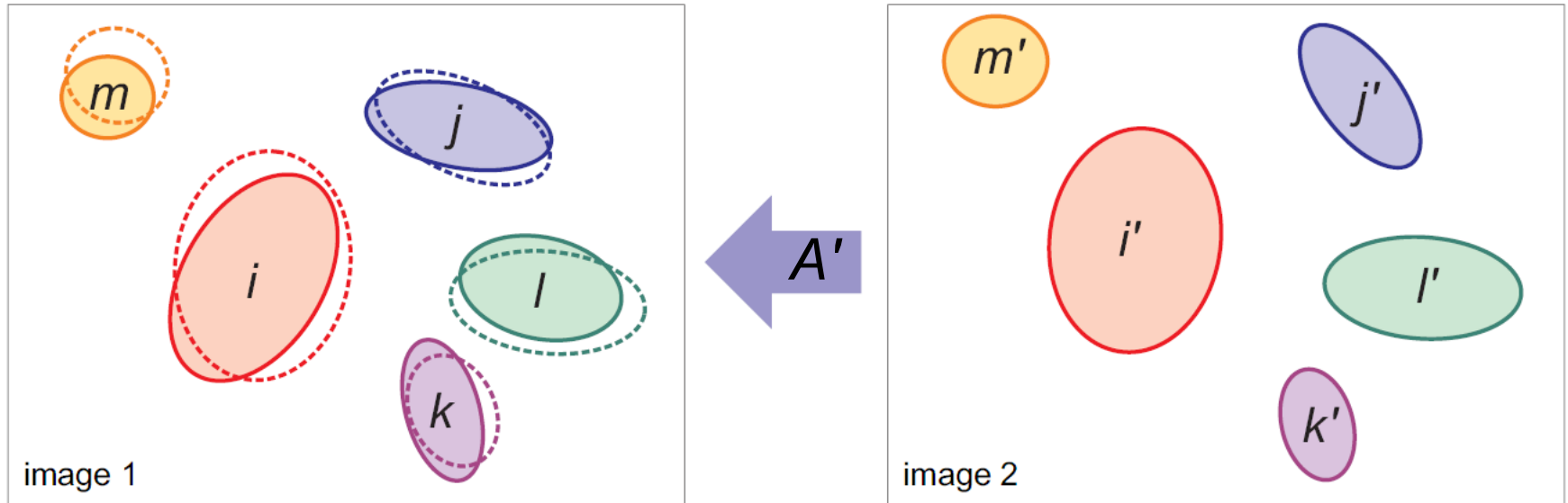


Beginning with each seed triple, repeat:

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Incremental alignment: Details

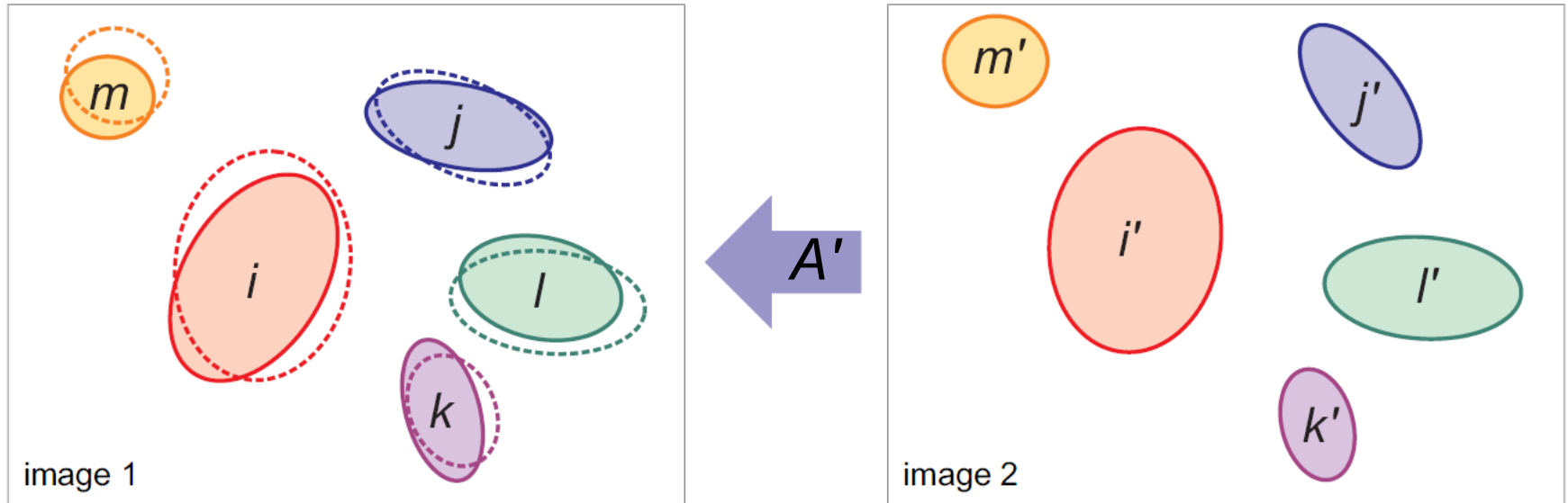


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Incremental alignment: Details



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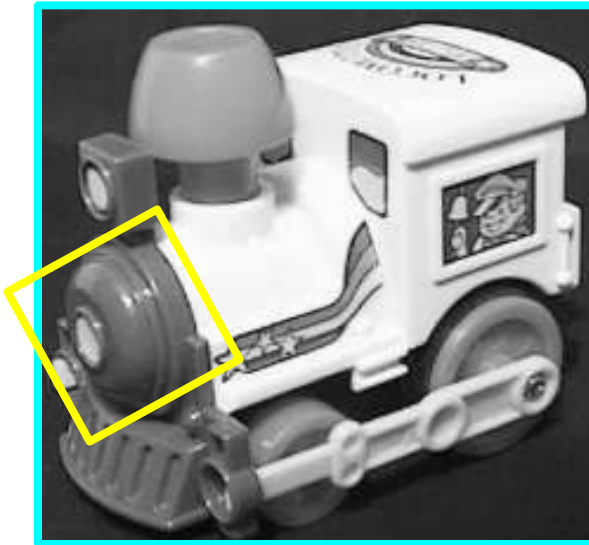
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Strategy 3: Hough transform

- Suppose our features are scale- and rotation-invariant
 - Then a single feature match provides an alignment hypothesis (translation, scale, orientation)

model

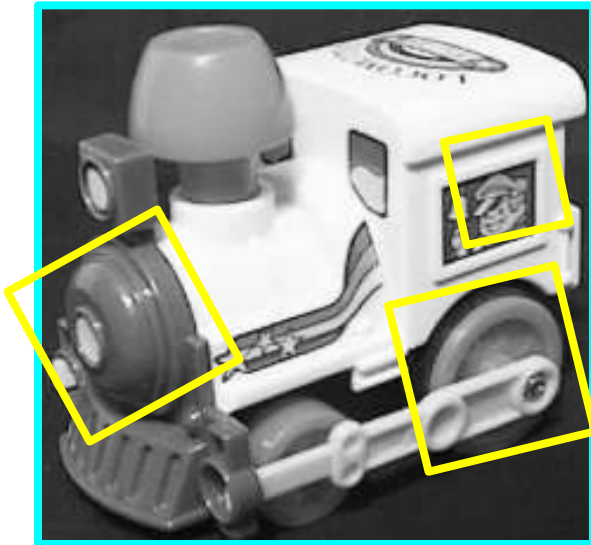


David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#)
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Strategy 3: Hough transform

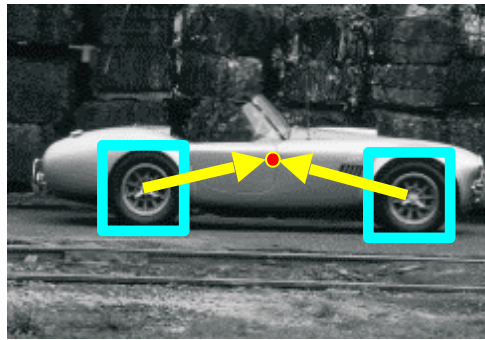
- Suppose our features are scale- and rotation-invariant
 - Then a single feature match provides an alignment hypothesis (translation, scale, orientation)
 - Of course, a hypothesis obtained from a single match is unreliable
 - Solution: let each match vote for its hypothesis in a Hough space with very coarse bins

model

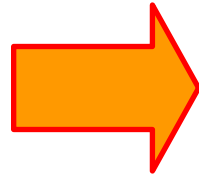


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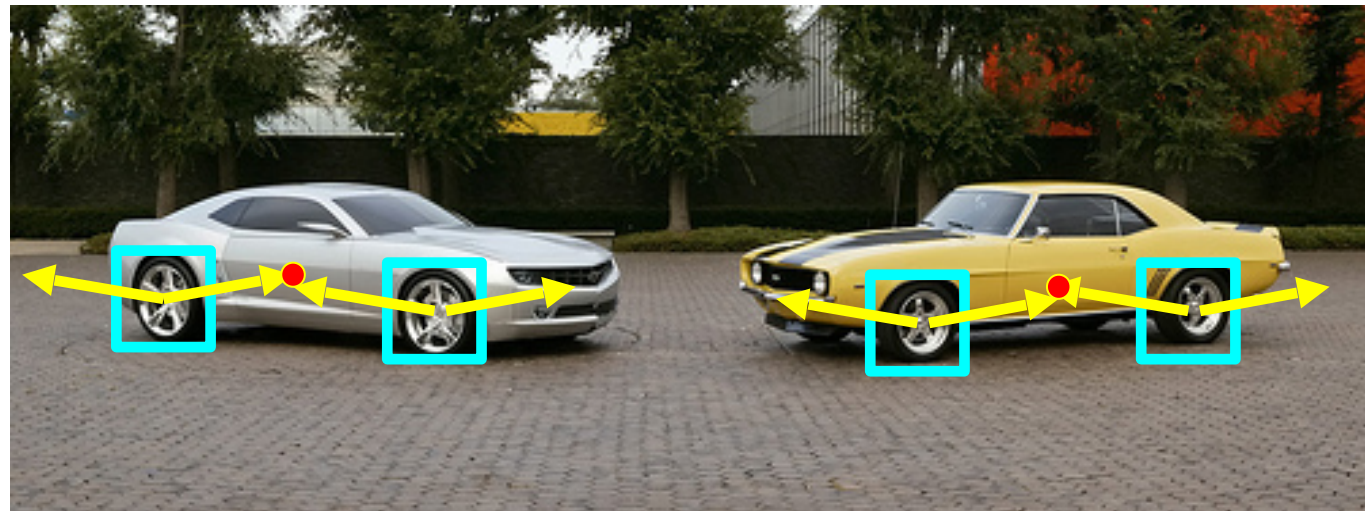
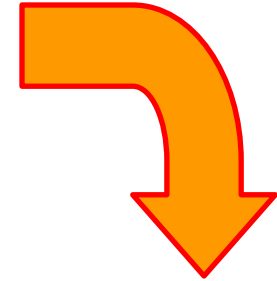
Recall: Generalized Hough transform



model



visual codeword with
displacement vectors



test image

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Hough transform details (D. Lowe's system)

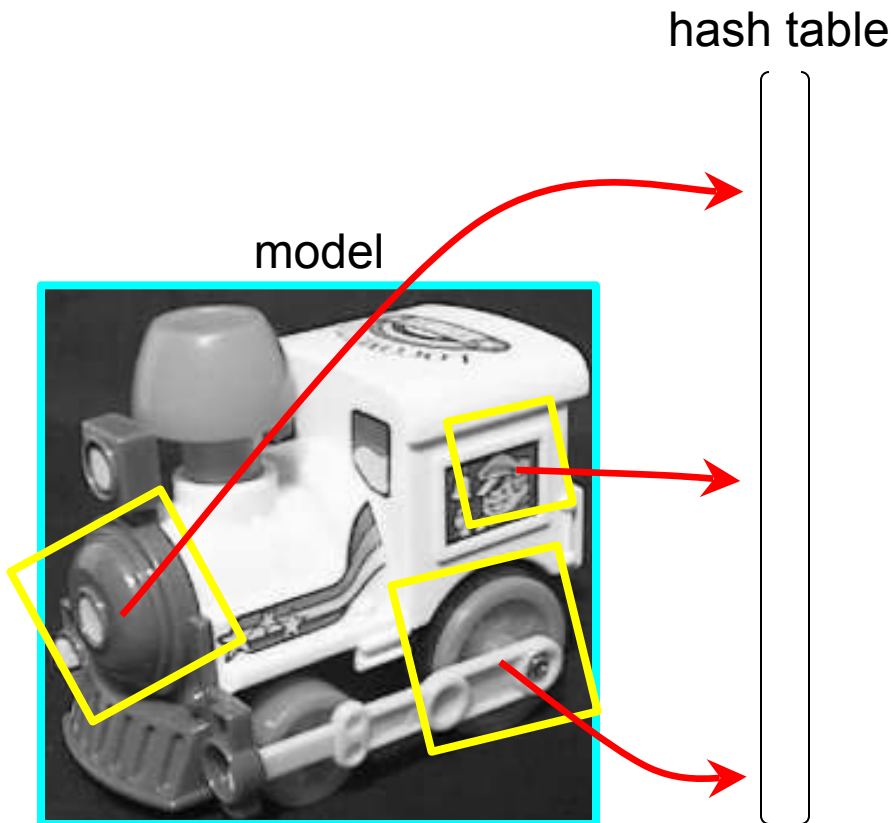
- **Training phase:** For each model feature, record 2D location, scale, and orientation of model (relative to normalized feature frame)
- **Test phase:** Let each match between a test and a model feature vote in a 4D Hough space
 - Use broad bin sizes of 30 degrees for orientation, a factor of 2 for scale, and 0.25 times image size for location
 - Vote for two closest bins in each dimension
- Find all bins with at least three votes and perform geometric verification
 - Estimate least squares *affine* transformation
 - Use stricter thresholds on transformation residual
 - Search for additional features that agree with the alignment

Image alignment: Review

- What is the bias/variance tradeoff?
- What are the advantages of a 2D affine alignment model?
- What are the main steps in feature-based alignment?
- How do we generate putative matches?
- How does the SIFT descriptor work?
- What are different computational strategies for estimating an alignment in the presence of outliers?
 - Strategy 1: RANSAC
 - Strategy 2: Incremental alignment
 - Strategy 3: Hough transform
 - Strategy 4: Hashing (next)

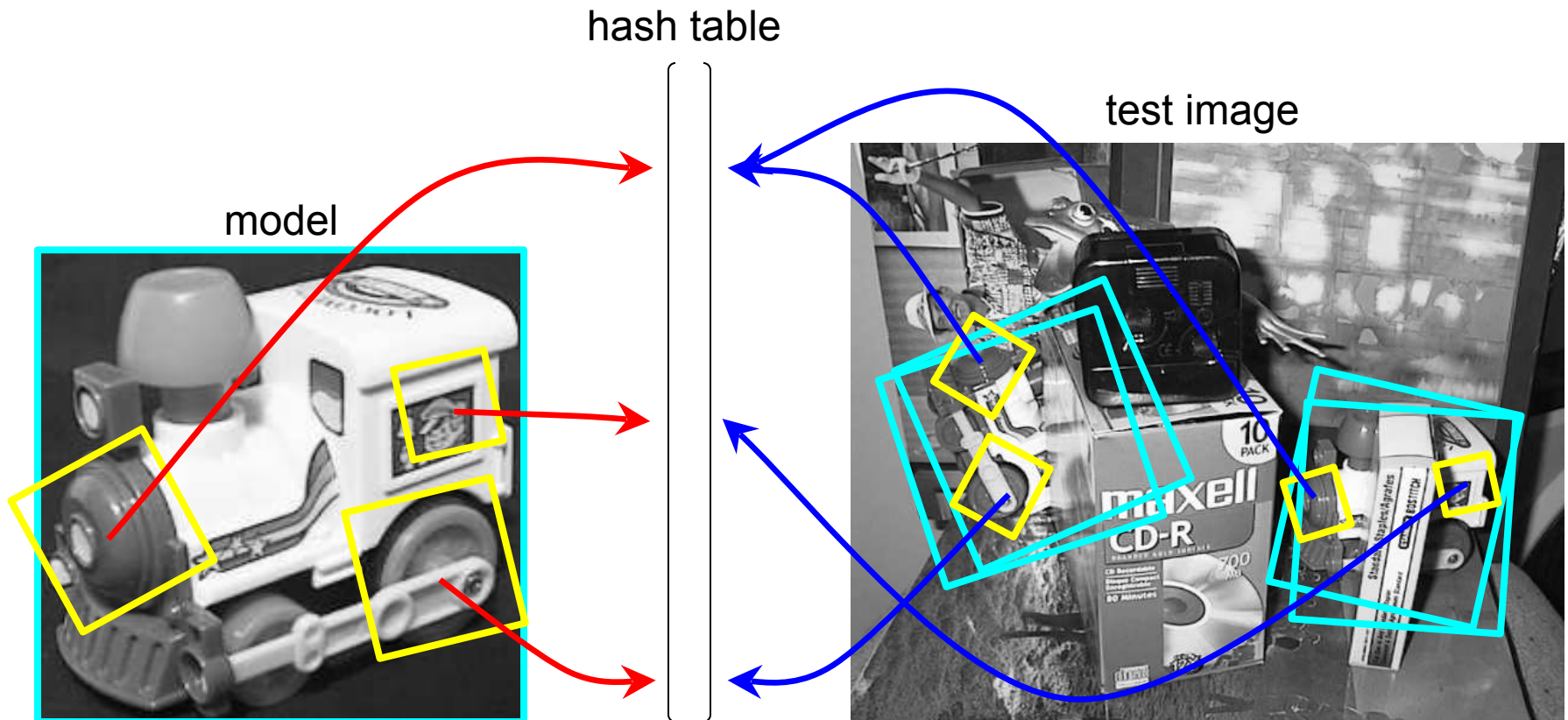
Strategy 4: Hashing

- Make each invariant image feature into a low-dimensional “key” that indexes into a table of hypotheses



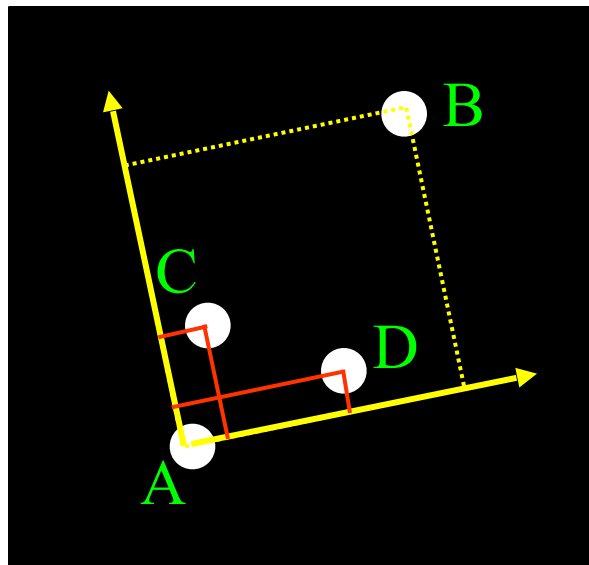
Strategy 4: Hashing

- Make each invariant image feature into a low-dimensional “key” that indexes into a table of hypotheses
- Given a new test image, compute the hash keys for all features found in that image, access the table, and look for consistent hypotheses



Strategy 4: Hashing

- Make each invariant image feature into a low-dimensional “key” that indexes into a table of hypotheses
- Given a new test image, compute the hash keys for all features found in that image, access the table, and look for consistent hypotheses
- This can even work when we don’t have any feature descriptors: we can take n-tuples of neighboring features and compute invariant hash codes from their geometric configurations



Application: Searching the sky

<http://www.astrometry.net/>

Beyond affine transformations

- What is the transformation between two views of a planar surface?

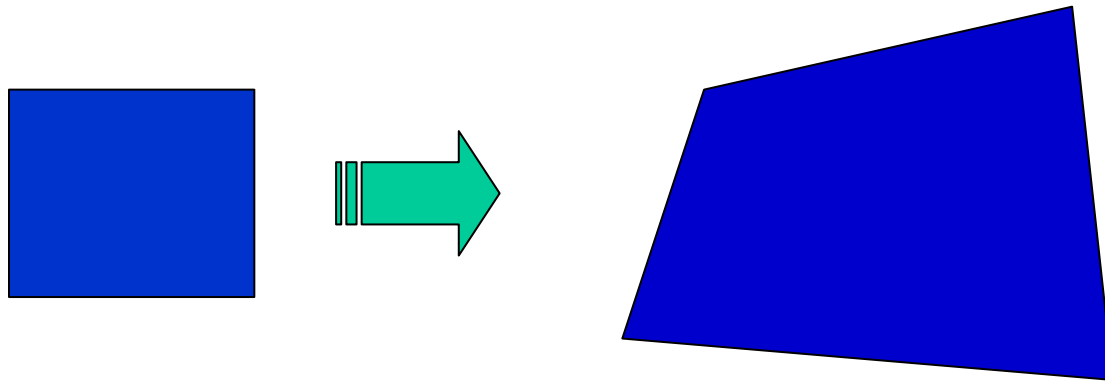


- What is the transformation between images from two cameras that share the same center?



Beyond affine transformations

- **Homography:** plane projective transformation (transformation taking a quad to another arbitrary quad)



Fitting a homography

- Recall: homogenous coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Converting *to* homogenous
image coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

Converting *from* homogenous
image coordinates

Fitting a homography

- Recall: homogenous coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Converting *to* homogenous
image coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

Converting *from* homogenous
image coordinates

- Equation for homography:

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Fitting a homography

- Equation for homography:

$$\lambda \begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \qquad \lambda \mathbf{x}'_i = \mathbf{H} \mathbf{x}_i = \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \\ \mathbf{h}_3^T \end{bmatrix} \mathbf{x}_i$$

9 entries, 8 degrees of freedom
(scale is arbitrary)

$$\mathbf{x}'_i \times \mathbf{H} \mathbf{x}_i = 0$$

$$\mathbf{x}'_i \times \mathbf{H} \mathbf{x}_i = \begin{bmatrix} y'_i \mathbf{h}_3^T \mathbf{x}_i - \mathbf{h}_2^T \mathbf{x}_i \\ \mathbf{h}_1^T \mathbf{x}_i - x'_i \mathbf{h}_3^T \mathbf{x}_i \\ x'_i \mathbf{h}_2^T \mathbf{x}_i - y'_i \mathbf{h}_1^T \mathbf{x}_i \end{bmatrix}$$

$$\begin{bmatrix} 0^T & -\mathbf{x}_i^T & y'_i \mathbf{x}_i^T \\ \mathbf{x}_i^T & 0^T & -x'_i \mathbf{x}_i^T \\ -y'_i \mathbf{x}_i^T & x'_i \mathbf{x}_i^T & 0^T \end{bmatrix} \begin{pmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{pmatrix} = 0$$

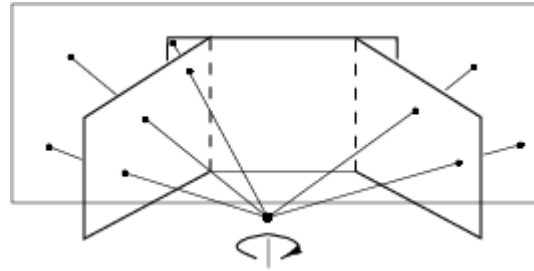
3 equations, only 2 linearly independent

Direct linear transform

$$\begin{bmatrix} 0^T & \mathbf{x}_1^T & -y'_1 \mathbf{x}_1^T \\ \mathbf{x}_1^T & 0^T & -x'_1 \mathbf{x}_1^T \\ \boxed{?} & \boxed{?} & \boxed{?} \\ 0^T & \mathbf{x}_n^T & -y'_n \mathbf{x}_n^T \\ \mathbf{x}_n^T & 0^T & -x'_n \mathbf{x}_n^T \end{bmatrix} \begin{pmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{pmatrix} = 0 \quad \mathbf{A} \mathbf{h} = 0$$

- H has 8 degrees of freedom (9 parameters, but scale is arbitrary)
- One match gives us two linearly independent equations
- Four matches needed for a minimal solution (null space of 8x9 matrix)
- More than four: homogeneous least squares

Application: Panorama stitching



Recognizing panoramas

- Given contents of a camera memory card, automatically figure out which pictures go together and stitch them together into panoramas

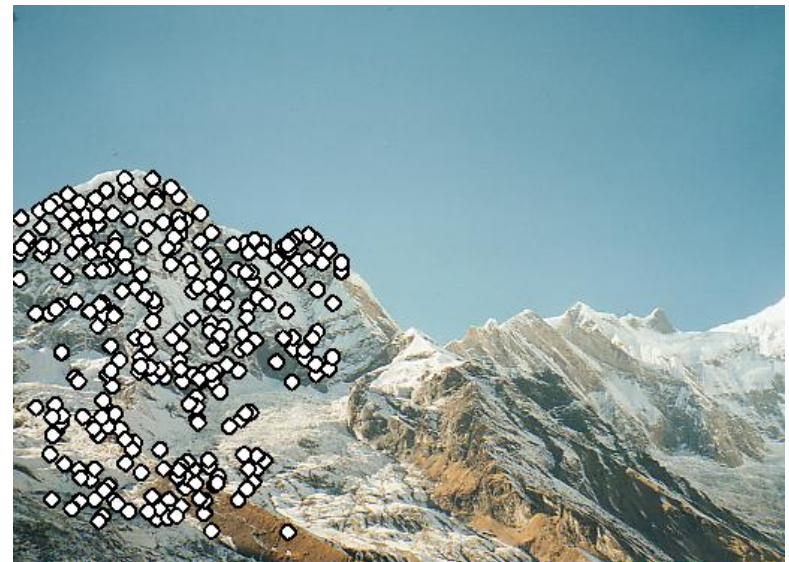
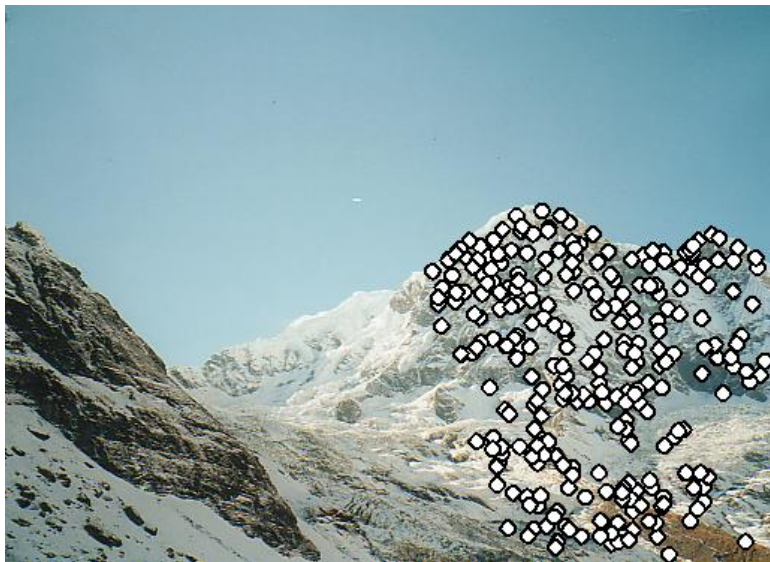


M. Brown and D. Lowe, ["Recognizing Panoramas."](http://www.cs.ubc.ca/~mbrown/panorama/panorama.html) ICCV 2003.
<http://www.cs.ubc.ca/~mbrown/panorama/panorama.html>

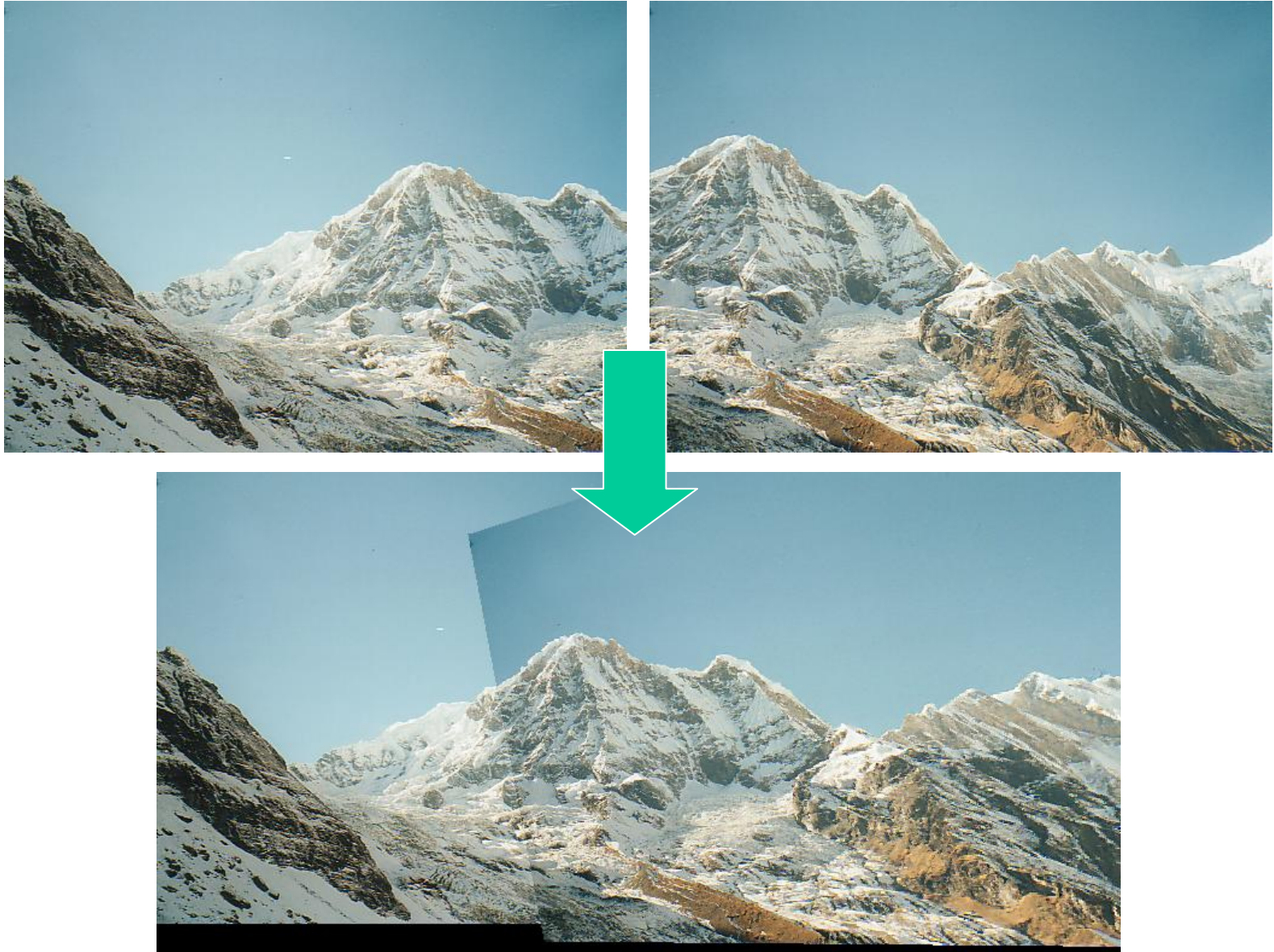
1. Estimate homography (RANSAC)



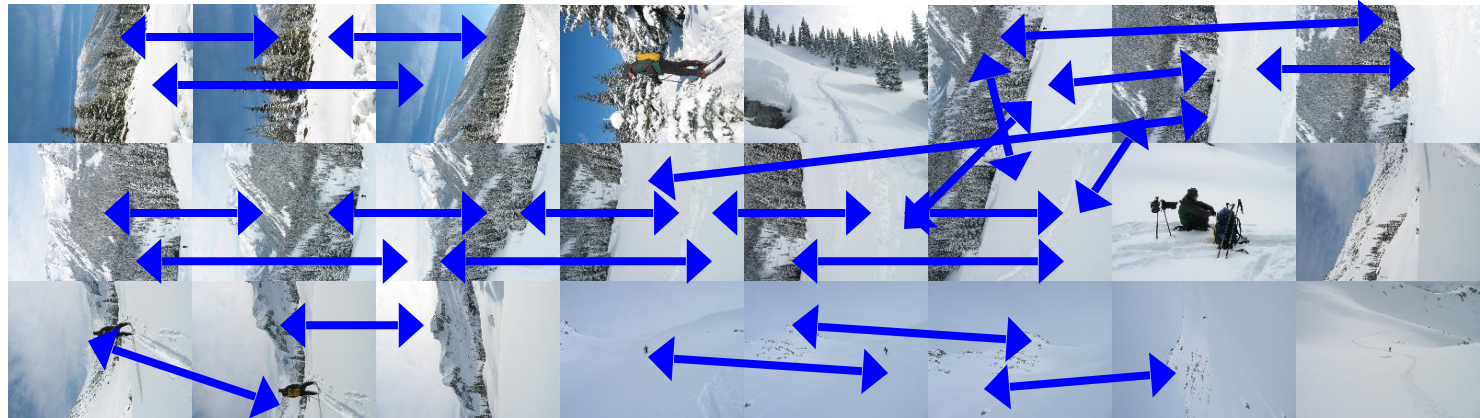
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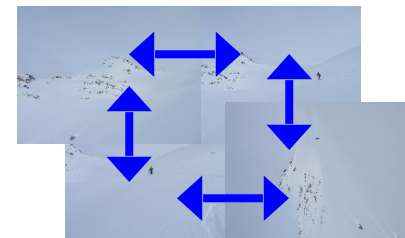
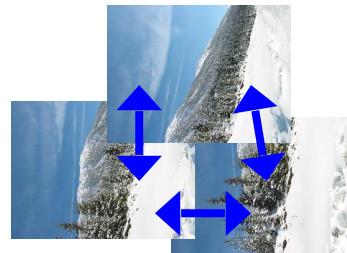
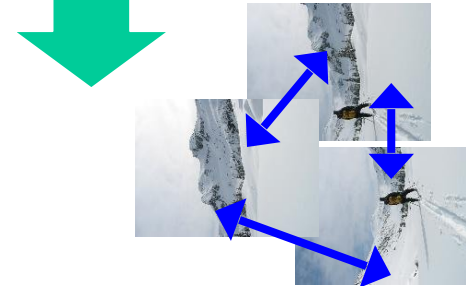
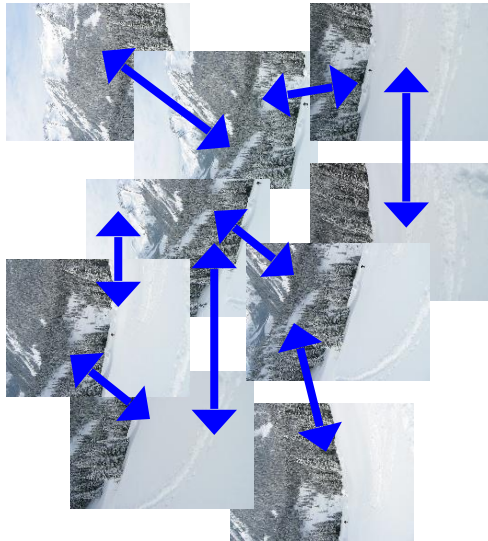
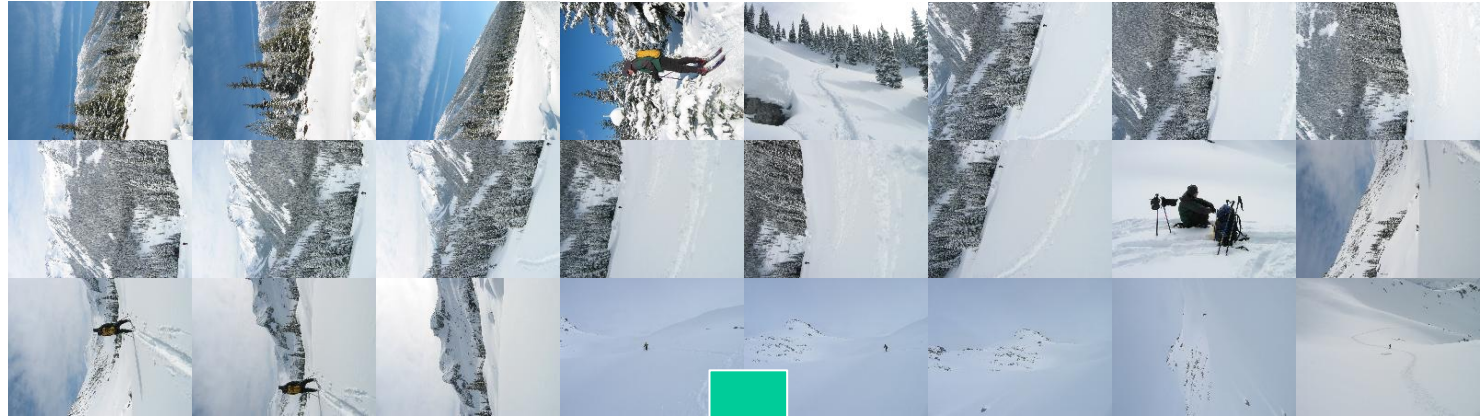
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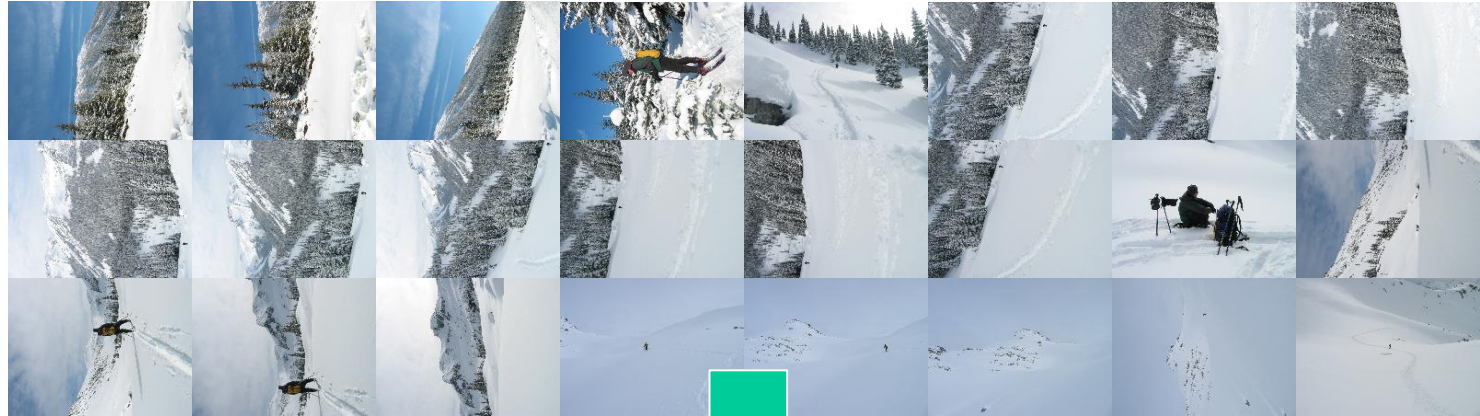
2. Find connected sets of images



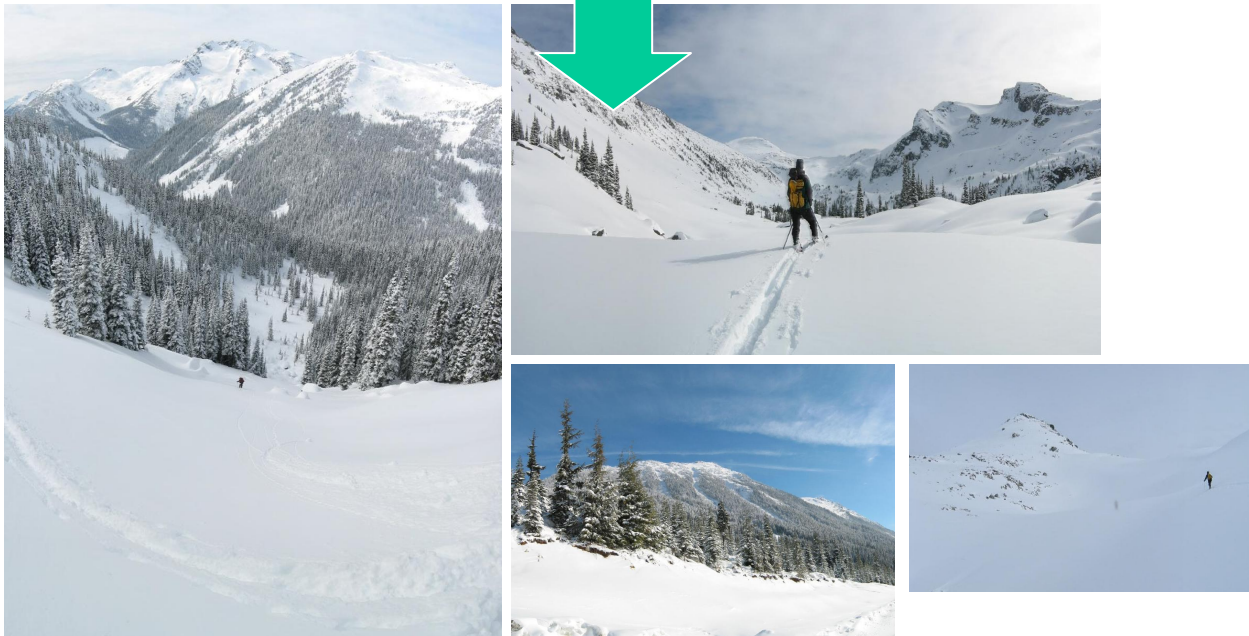
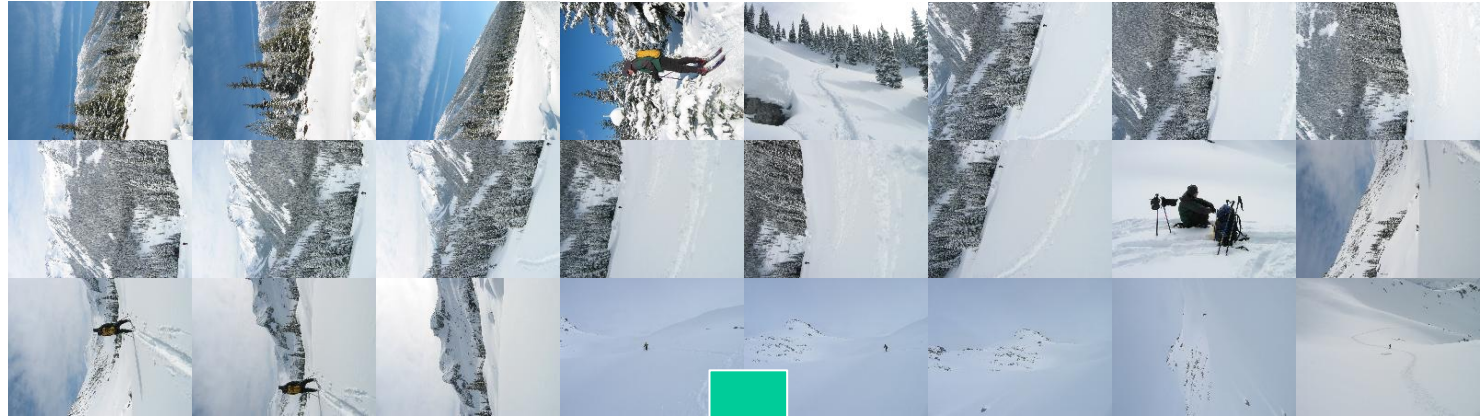
2. Find connected sets of images



2. Find connected sets of images



3. Stitch and blend the panoramas



Results



Issues in alignment-based applications

- Choosing the geometric alignment model
 - Tradeoff between “correctness” and robustness (also, efficiency)
- Choosing the descriptor
 - “Rich” imagery (natural images): high-dimensional patch-based descriptors (e.g., SIFT)
 - “Impoverished” imagery (e.g., star fields): need to create invariant geometric descriptors from k-tuples of point-based features
- Strategy for finding putative matches
 - Small number of images, one-time computation (e.g., panorama stitching): brute force search
 - Large database of model images, frequent queries: indexing or hashing
 - Heuristics for feature-space pruning of putative matches

Issues in alignment-based applications

- Choosing the geometric alignment model
- Choosing the descriptor
- Strategy for finding putative matches
- Hypothesis generation strategy
 - Relatively large inlier ratio: RANSAC
 - Small inlier ratio: locality constraints, Hough transform
- Hypothesis verification strategy
 - Size of consensus set, residual tolerance depend on inlier ratio and expected accuracy of the model
 - Possible refinement of geometric model
 - Dense verification

Next time: Single-view geometry

